

Data Engineering 101: DATABASES



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DATABASE MANAGEMENT SYSTEM (DBMS) EXPLANATION

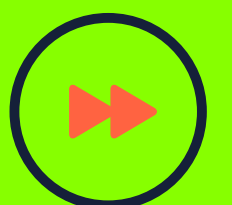
*A software system that uses a
standard method to store and
organize data.*

EXAMPLE

MySQL, PostgreSQL



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STORAGE ENGINE

EXPLANATION

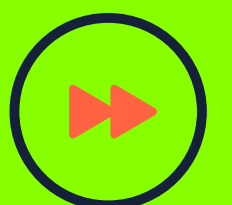
A component of DBMS responsible for storing, retrieving, and managing data in memory and on

EXAMPLE

InnoDB, RocksDB



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MEMORY-BASED DBMS

EXPLANATION

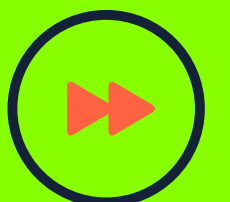
Databases that store data primarily in memory, using disk for recovery and logging.

EXAMPLE

Redis



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DISK-BASED DBMS

EXPLANATION

Databases that hold most of the data on disk, using memory for caching and temporary storage.

EXAMPLE

PostgreSQL



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DURABILITY IN MEMORY-BASED STORES

EXPLANATION

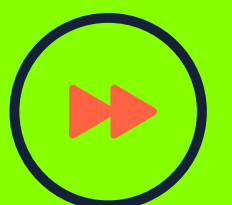
Ensuring data persistence in memory-based DBMS by maintaining backups on disk.

EXAMPLE

Redis with AOF persistence



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ROW-ORIENTED DBMS

EXPLANATION

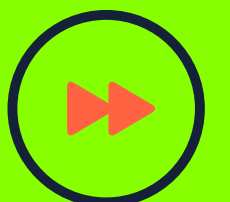
Databases that store data in rows, optimized for OLTP workloads.

EXAMPLE

MySQL, PostgreSQL



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COLUMN-ORIENTED DBMS

EXPLANATION

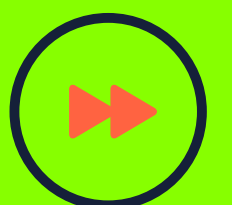
Databases that store data in columns, optimized for OLAP workloads.

EXAMPLE

Apache Cassandra, HBase



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DATA FILES

EXPLANATION

Files that store data records in DBMS.

EXAMPLE

MySQL's .ibd files



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INDEX FILES

EXPLANATION

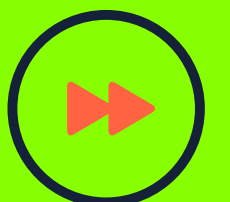
Files that store metadata to locate records in data files efficiently.

EXAMPLE

MySQL's .frm files



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PRIMARY INDEX

EXPLANATION

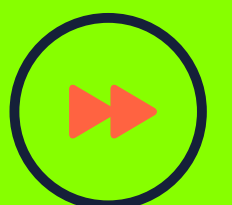
An index that determines the primary key for accessing data records.

EXAMPLE

Primary key in a MySQL table



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SECONDARY INDEX

EXPLANATION

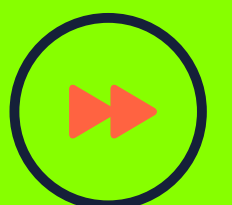
An index that facilitates access by keys other than the primary one.

EXAMPLE

Index on a non-primary key column



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HEAP FILES

EXPLANATION

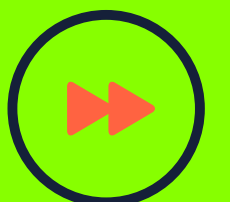
Data files where records are stored in no particular order.

EXAMPLE

MySQL MyISAM storage engine



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HASHED FILES

EXPLANATION

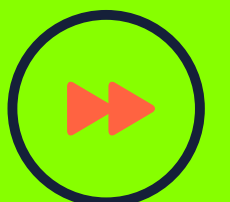
Data files where records are stored based on a hash value of the key.

EXAMPLE

Hash index in PostgreSQL



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INDEX-ORGANIZED TABLES (IOT)

EXPLANATION

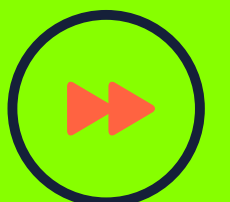
Tables where data records are stored in the index itself.

EXAMPLE

Oracle IOT



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CLUSTERING

EXPLANATION

Storing data records in index order to improve query performance.

EXAMPLE

Clustered index in SQL Server



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NON-CLUSTERING

EXPLANATION

Storing data records without following the index order.

EXAMPLE

Non-clustered index in SQL Server



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BUFFERING

EXPLANATION

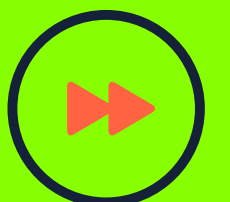
Collecting a certain amount of data in memory before writing it to disk.

EXAMPLE

Write buffer in RocksDB



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IMMUTABILITY

EXPLANATION

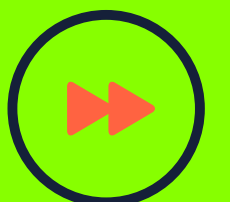
Using immutable files for storing data, preventing modifications.

EXAMPLE

Write-once logs in Apache Kafka



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ORDERING

EXPLANATION

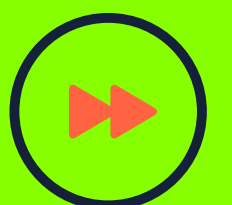
Storing values in a specific order to optimize access patterns.

EXAMPLE

Sorted strings in LSM trees



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B-TREES

EXPLANATION

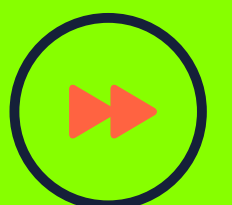
A balanced tree data structure that maintains sorted data and allows searches, sequential access, insertions, and deletions.

EXAMPLE

B-Tree indexes in MySQL



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B-TREE VARIANTS

EXPLANATION

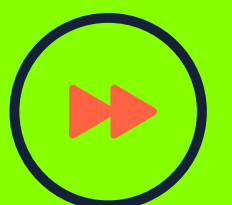
Different implementations of B-Trees to optimize for specific use cases.

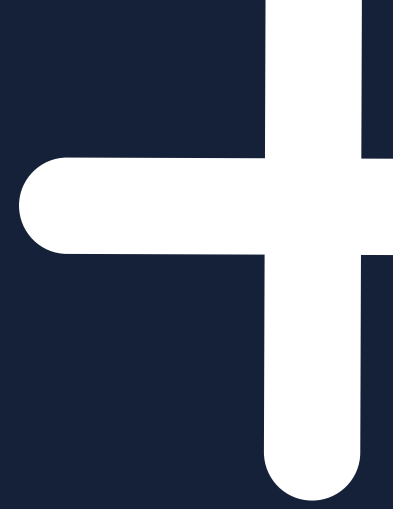
EXAMPLE

B+ Trees, B Trees*



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LOG-STRUCTURED STORAGE

EXPLANATION

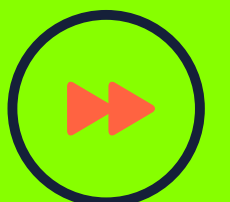
A storage model that logs all changes sequentially to optimize write performance.

EXAMPLE

LSM trees in Cassandra



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LSM TREES

EXPLANATION

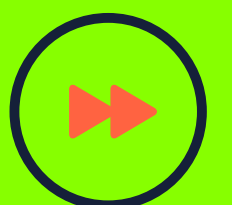
A type of log-structured storage optimized for write-heavy workloads.

EXAMPLE

LevelDB, RocksDB



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SORTED STRING TABLES (SSTABLES)

EXPLANATION

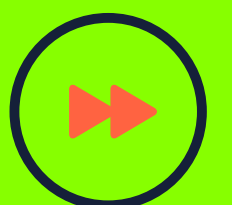
Immutable data files used in LSM trees, containing sorted key-value pairs.

EXAMPLE

SSTables in Cassandra



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BLOOM FILTERS

EXPLANATION

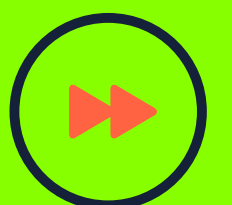
A space-efficient probabilistic data structure used to test whether an element is in a set.

EXAMPLE

Bloom filters in Cassandra



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SKIPLIST

EXPLANATION

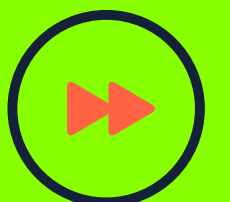
A data structure that allows fast search within an ordered sequence of elements.

EXAMPLE

Skiplist in Redis



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BUFFER MANAGER

EXPLANATION

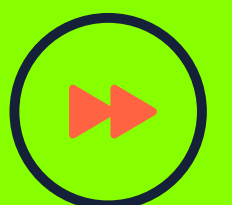
A component that manages data caching in memory.

EXAMPLE

Buffer cache in PostgreSQL



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RECOVERY MANAGER

EXPLANATION

A component responsible for restoring the system state after failures.

EXAMPLE

WAL in PostgreSQL



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CONCURRENCY CONTROL

EXPLANATION

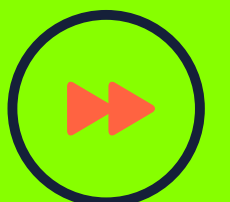
Mechanisms to ensure correct results for concurrent operations.

EXAMPLE

MVCC in PostgreSQL



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TRANSACTION ISOLATION

EXPLANATION

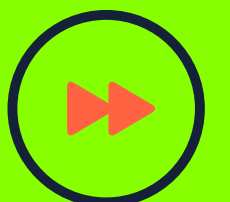
The degree to which the operations in one transaction are isolated from those in other transactions.

EXAMPLE

Serializable, Read Committed isolation levels



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OPTIMISTIC CONCURRENCY CONTROL EXPLANATION

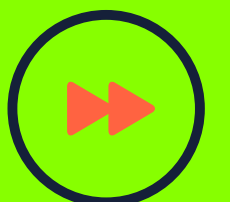
A method where transactions execute without locking resources and check for conflicts before commit.

EXAMPLE

OCC in Oracle



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PESSIMISTIC CONCURRENCY CONTROL EXPLANATION

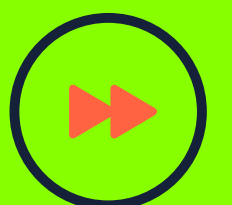
A method where transactions lock resources before accessing them to prevent conflicts.

EXAMPLE

Lock-based transactions in MySQL



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LOCK-BASED CONCURRENCY CONTROL EXPLANATION

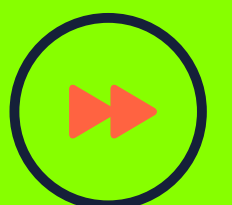
A method that uses locks to manage concurrent access to data.

EXAMPLE

Two-phase locking in SQL Server



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MULTIVERSION CONCURRENCY CONTROL (MVCC) EXPLANATION

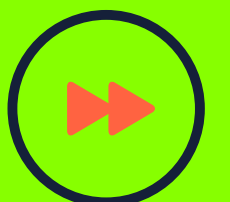
A concurrency control method that keeps multiple versions of data to increase transaction isolation.

EXAMPLE

MVCC in PostgreSQL



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TWO-PHASE COMMIT (2PC)

EXPLANATION

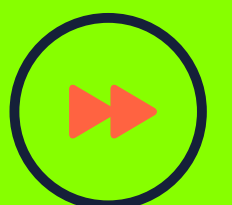
A protocol to ensure all participants in a distributed transaction agree to commit or abort.

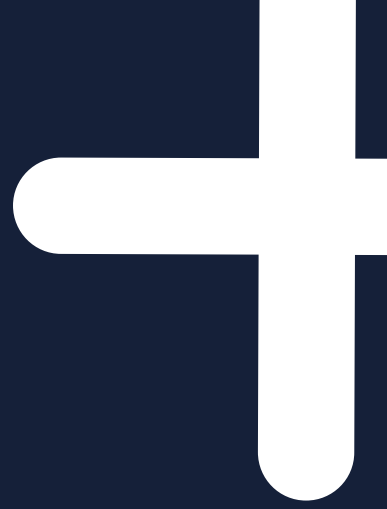
EXAMPLE

2PC in distributed databases



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THREE-PHASE COMMIT (3PC)

EXPLANATION

An extension of 2PC to reduce the chances of blocking in case of failures.

EXAMPLE

3PC in distributed databases



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PAXOS

EXPLANATION

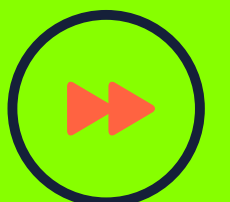
A consensus algorithm to achieve agreement among distributed systems.

EXAMPLE

Used in Google Spanner



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RAFT

EXPLANATION

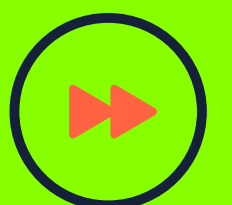
A consensus algorithm designed for managing replicated logs.

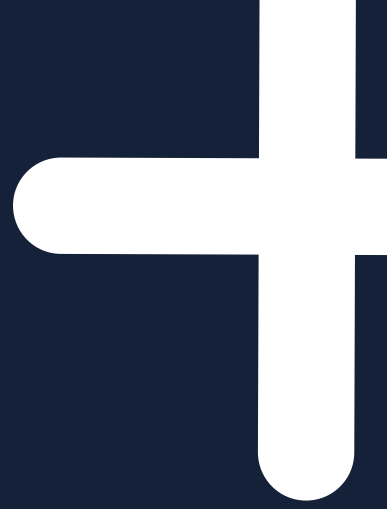
EXAMPLE

Used in etcd, Consul



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BYZANTINE FAULT TOLERANCE (BFT)

EXPLANATION

A property of systems that can tolerate Byzantine faults.

EXAMPLE

PBFT algorithm



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LOG STACKING

EXPLANATION

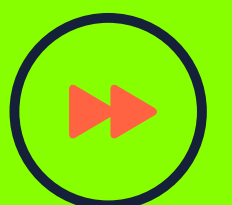
Organizing multiple logs for efficiency and reliability.

EXAMPLE

LLAMA in distributed systems



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FLASH TRANSLATION LAYER (FTL)

EXPLANATION

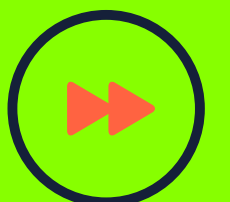
A layer that manages flash memory operations, translating logical block addresses to physical addresses.

EXAMPLE

FTL in SSDs



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CONSISTENT HASHING

EXPLANATION

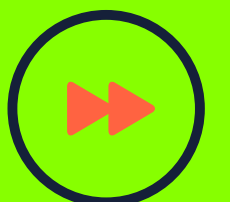
A technique to distribute data across multiple nodes to minimize reorganization when nodes are added or removed.

EXAMPLE

Used in DynamoDB



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DISTRIBUTED SYSTEMS

ABSTRACTIONS

EXPLANATION

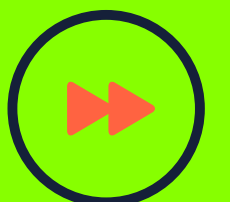
Concepts to simplify the design and implementation of distributed systems.

EXAMPLE

CAP theorem



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LEADER ELECTION

EXPLANATION

Algorithms to elect a leader in a distributed system to coordinate tasks.

EXAMPLE

Bully algorithm



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FAILURE DETECTION

EXPLANATION

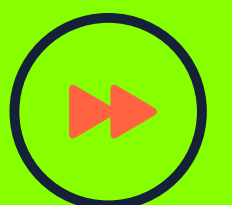
Mechanisms to identify failures in distributed systems.

EXAMPLE

Heartbeats and pings



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GOSSIP PROTOCOL

EXPLANATION

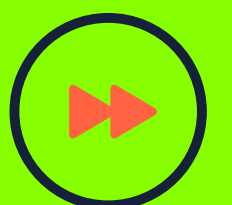
A communication protocol to spread information in distributed systems.

EXAMPLE

Gossip protocol in Cassandra



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ANTI-ENTROPY

EXPLANATION

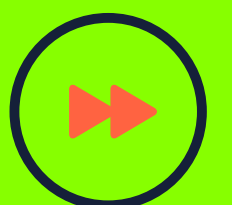
Techniques to reconcile differences between replicas in distributed systems.

EXAMPLE

Merkle trees



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DIGEST READS

EXPLANATION

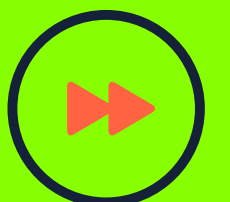
A method to check data consistency by comparing digests of data.

EXAMPLE

Digest reads in Cassandra



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HINTED HANDOFF

EXPLANATION

A technique to handle temporary failures in distributed systems by storing write operations for later replay.

EXAMPLE

Hinted handoff in Cassandra



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MERKLE TREES

EXPLANATION

A tree structure for efficient data verification and synchronization.

EXAMPLE

Used in Git and Cassandra



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OVERLAY NETWORKS

EXPLANATION

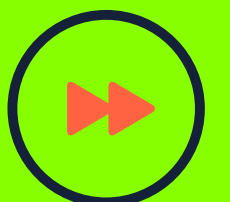
Networks built on top of another network to provide additional features.

EXAMPLE

Virtual networks in cloud services



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ATOMIC BROADCAST

EXPLANATION

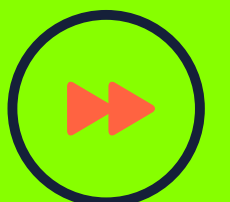
Ensuring messages are delivered in the same order across all nodes in a distributed system.

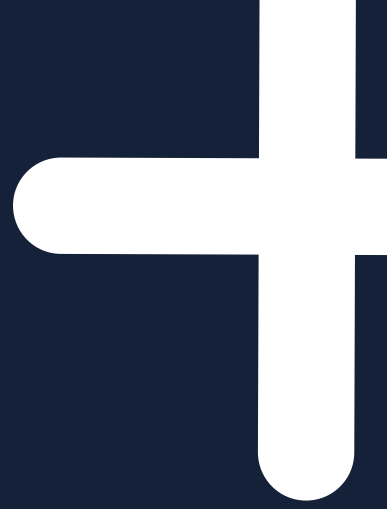
EXAMPLE

ZAB in ZooKeeper



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VIRTUAL SYNCHRONY

EXPLANATION

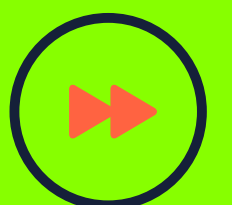
A model to ensure consistent group membership and message delivery order.

EXAMPLE

Used in ISIS2 library



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TWO GENERALS' PROBLEM

EXPLANATION

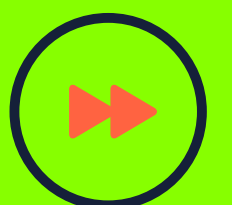
A thought experiment illustrating the challenges of coordination in distributed systems.

EXAMPLE

Problem statement in distributed computing



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FLP IMPOSSIBILITY

EXPLANATION

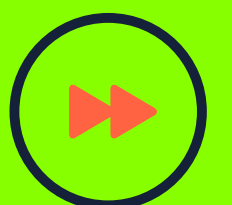
A theorem stating that consensus is impossible in an asynchronous system with one faulty process.

EXAMPLE

FLP result in distributed systems



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SYSTEM SYNCHRONY

EXPLANATION

The assumption about the bounds on message delays and clock drifts in a distributed system.

EXAMPLE

Synchronous, asynchronous systems



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FAILURE MODELS

EXPLANATION

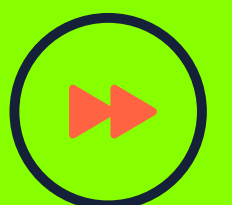
Different ways to categorize and handle failures in distributed systems.

EXAMPLE

*Crash faults, omission faults,
Byzantine faults*



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CAP THEOREM

EXPLANATION

The principle that a distributed system can only guarantee two out of three: Consistency, Availability, Partition tolerance.

EXAMPLE

CAP trade-offs in NoSQL databases



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EVENTUAL CONSISTENCY

EXPLANATION

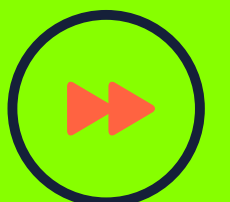
A consistency model where updates will propagate to all nodes eventually.

EXAMPLE

Eventual consistency in DynamoDB



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STRONG CONSISTENCY

EXPLANATION

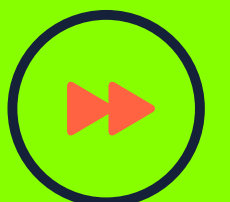
A consistency model ensuring immediate visibility of updates to all nodes.

EXAMPLE

Linearizability



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CAUSAL CONSISTENCY

EXPLANATION

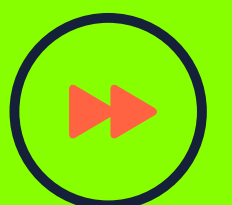
A consistency model ensuring that operations that are causally related are seen in the correct order.

EXAMPLE

Causal consistency in distributed systems



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LINEARIZABILITY

EXPLANATION

A strong consistency model where operations appear to occur instantaneously.

EXAMPLE

Linearizability in databases



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SEQUENTIAL CONSISTENCY

EXPLANATION

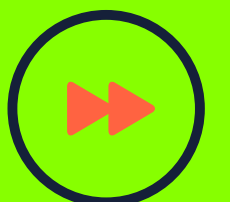
A consistency model where the results of operations are the same as if they were executed in some sequential order.

EXAMPLE

Sequential consistency in memory models



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CRDTS (CONFLICT-FREE REPLICATED DATA TYPES)

EXPLANATION

Data structures designed to ensure strong eventual consistency.

EXAMPLE

CRDTs in distributed systems



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HARVEST AND YIELD

EXPLANATION

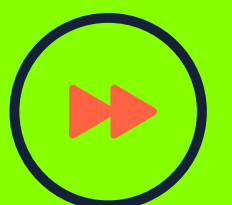
A model to understand the trade-offs between availability and consistency.

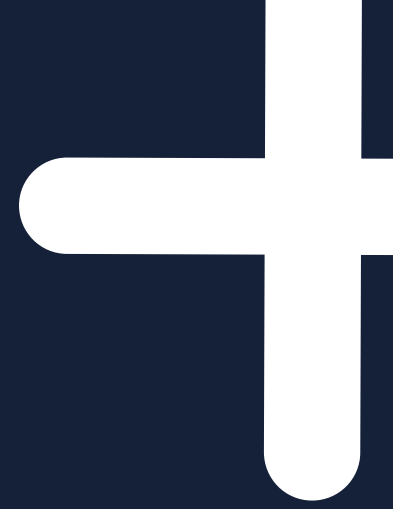
EXAMPLE

Used in database design



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DISTRIBUTED TRANSACTIONS

EXPLANATION

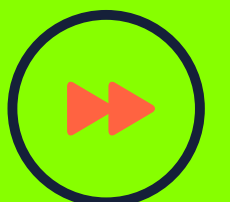
Transactions that span multiple nodes or databases.

EXAMPLE

Two-phase commit, three-phase commit



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COORDINATION AVOIDANCE

EXPLANATION

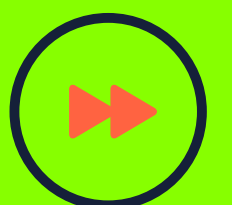
Techniques to reduce the need for coordination in distributed transactions.

EXAMPLE

Coordination-free algorithms



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WITNESS REPLICAS

EXPLANATION

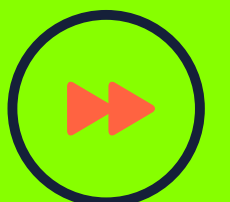
Lightweight replicas used to improve availability and consistency.

EXAMPLE

Witness nodes in database clusters



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CONSISTENCY MODELS

EXPLANATION

The guarantees provided by a database system regarding the visibility and ordering of updates.

EXAMPLE

Strong, eventual, causal consistency



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DATA SHARDING

EXPLANATION

Splitting data across multiple databases to improve performance and scalability.

EXAMPLE

Sharding in MongoDB



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READ REPAIR

EXPLANATION

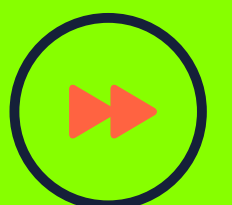
A technique to ensure data consistency during read operations.

EXAMPLE

Read repair in Cassandra



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STATE

CONSISTENCY

EXPLANATION

Ensuring all nodes in a distributed system have the same state.

EXAMPLE

Consistent state in distributed databases



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PROCESSING

EXPLANATION

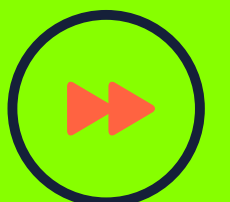
The execution of tasks in a distributed system.

EXAMPLE

Distributed processing frameworks



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CLOCKS AND TIME

EXPLANATION

The role of clocks and time synchronization in distributed systems.

EXAMPLE

NTP, logical clocks



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CASCADING FAILURES

EXPLANATION

A failure in one part of the system triggering failures in other parts.

EXAMPLE

Handling cascading failures in distributed systems



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HYBRID GOSSIP

EXPLANATION

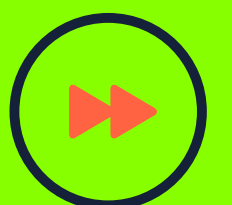
Combining gossip protocols with other communication methods.

EXAMPLE

Hybrid gossip in distributed systems



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PARTIAL VIEWS

EXPLANATION

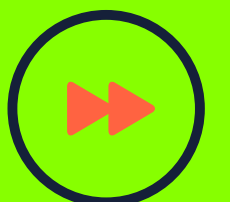
Maintaining a partial view of the network to improve scalability.

EXAMPLE

Partial views in peer-to-peer networks



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LOCAL AND REMOTE EXECUTION

EXPLANATION

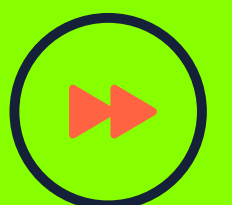
Executing tasks on local or remote nodes in a distributed system.

EXAMPLE

Remote procedure calls (RPCs)



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NETWORK PARTITIONS

EXPLANATION

Splitting the network into disjoint segments, affecting system availability.

EXAMPLE

Handling network partitions in distributed systems



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TRANSPARENT HANDLING OF FAILURES EXPLANATION

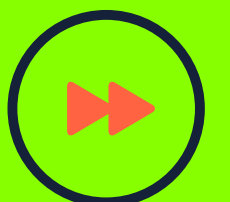
Designing systems to handle failures without user intervention.

EXAMPLE

Failover mechanisms in databases



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REVERSING FAILURE DETECTION PROBLEM STATEMENT

EXPLANATION

Techniques to ensure accurate failure detection in distributed systems.

EXAMPLE

Failure detection algorithms



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